Machine Learning for Time Series

Introduction

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Master MVA
2026-2027

Contents

1. Organization of the course

2. What is a time series ?

3. Data science for time series

4. Outline of the course

Organization of the course : lectures

- Six lectures on Monday morning at ENS Paris-Saclay
- Lectures will be in French but all material (slides, homeworks...) is in English.
- Attendance is mandatory **AND WILL BE CHECKED**. Unjustified absence = fail the class.
- Registration deadline for the mailing list : **October 8th, 2026**

METTRE LIEN

- ML for time series is part of the **Modelling track** !

Teaching material: http://www.laurentoudre.fr/ast.html

Organization of the course : tutorials

- For the tutorials there are two options :
  - Monday morning : on-site at ENS Paris-Saclay
  - Monday afternoon : remote on Zoom
- Content is identical in both slots: students should attend only one
- Extra work for each tutorial : approximately 6 hours
- Attendance is mandatory
- Tutorial homeworks are **mandatory**

missing or late homeworks → fail the class

Teaching assistant: Valerio Guerrini (valerio.guerrini@ens-paris-saclay.fr)

Validation

Validation: tutorials (25%) + mini-projects (25% report, 25% source code and 25% oral presentation)

- Projects can be done in groups of two, but no more than that
- Students are allowed to propose an additional project : ask in advance !
- The mini project consists of reading a research paper, implementing it in Python and running experiments on real time series
- Report (PDF file, ≈ 5 pages) + source code (Jupyter Notebook). A template for the report as well as detailed instructions will be sent.
- A 10 min oral presentation will finalize the course project
- **Due to the large number of students, there will be no auditeurs libres.**

Timeline

October 8th
- Registration deadline

October 19th–23rd
- Choice of mini-projects
  - Online form to select 5 mini-projects
  - Assignment giving one of their choices to as many students as possible
  - Students without a project contacted by email to choose another one

October 26th–30th
- Final assignment of mini-projects

December 13th–15th
- Evaluation: session 1
  - Report and code: December 13th (23:59)
  - Oral presentations: December 14th and 15th

January 3rd–6th
- Evaluation: session 2
  - Report and code: January 3rd (23:59)
  - Oral presentations: January 4th, 5th and 6th

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What is a time series?

- A time series is a series of data points **indexed in time order**
- In practice, array of real numbers of size $D \times N$ where D is the number of dimensions and N the number of samples
  - Sample number n

n	0	1	2	3	4	5	6

  - Time series values $x[n]$

$x[n]$							
	0.7	0.2	0.8	0.9	0.3	0.2	0.7
	0.4	0.1	0.6	0.2	0.5	0.6	0.3

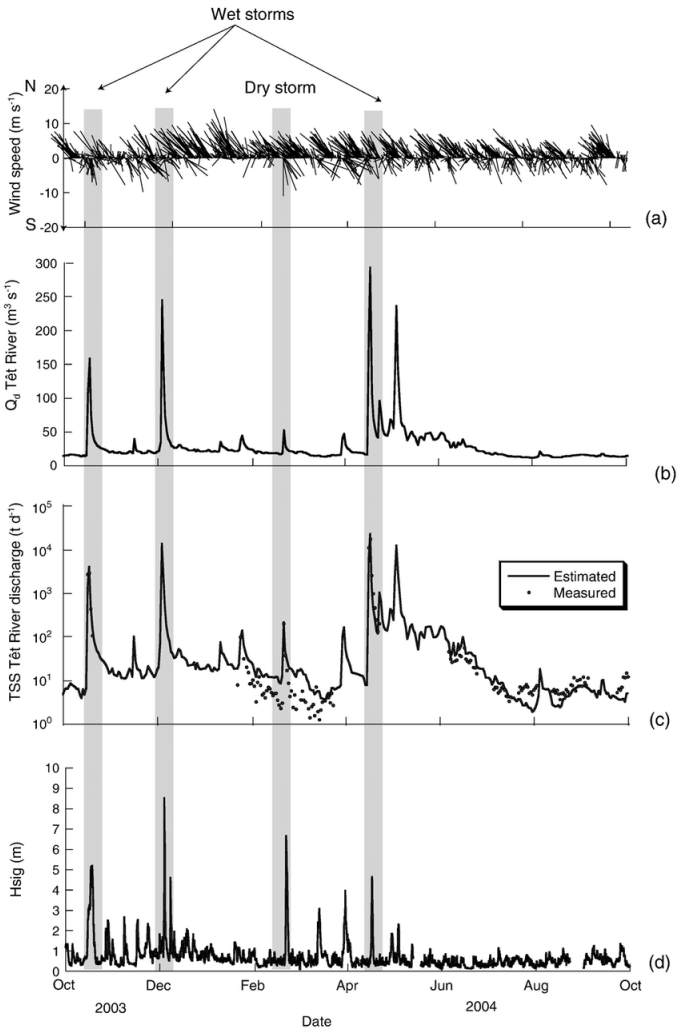
  - Timestamps $t[n]$

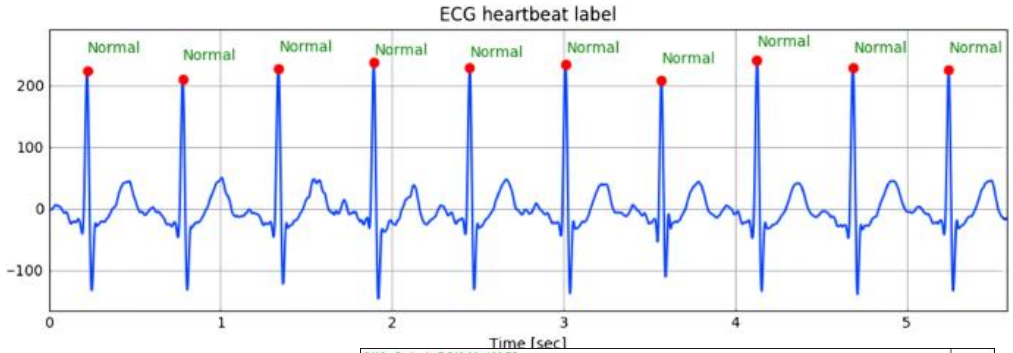
$t[n]$							
	16:30:01	16:30:23	16:31:43	16:32:38	16:33:06	16:33:16	16:33:56

An un-unified field

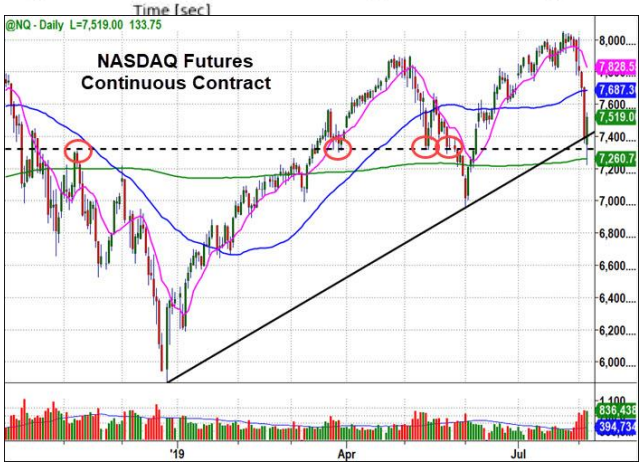
- Different scientific communities have given different names to the same mathematical object.
  - **Time series**: mathematics, statistics, economics, finance...
  - **Signals**: signal processing, physics, engineering, simulation...
  - **Sequences**: computer sciences, bioinformatics, data mining...
- In this course, we will use these terms **interchangeably**
- **Typical definition**: real-valued (or at least ordered) sequential data

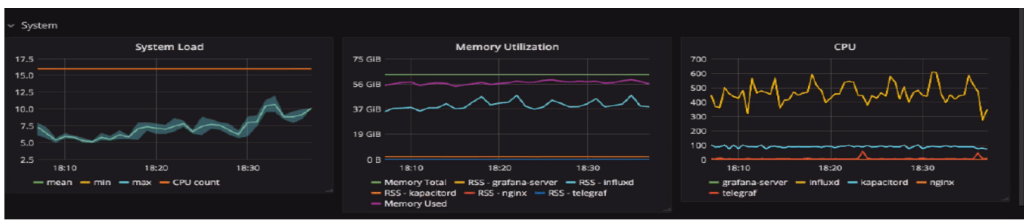
Time series are everywhere





Meteorology,
Finance,
Healthcare,
Monitoring,
Epidemiology,
Sensor networks...

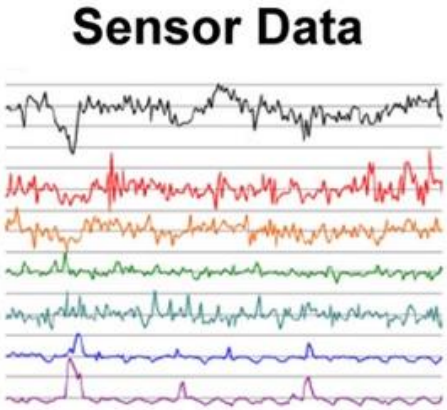


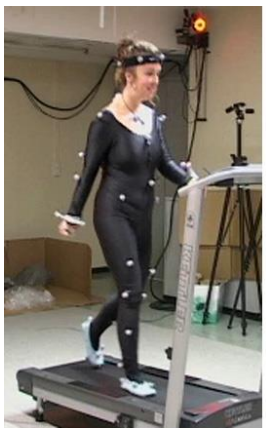


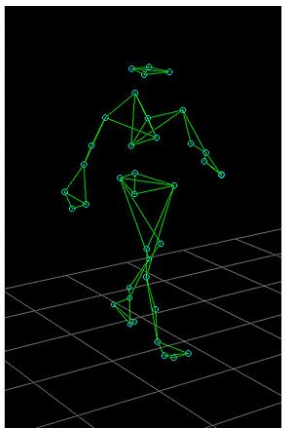
Univariate vs. multivariate

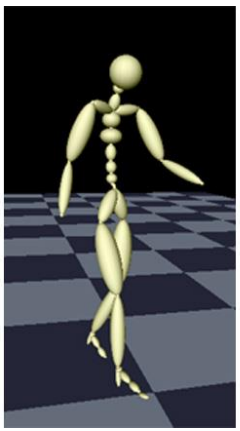


2D/3D trajectories,
Multivariate time series,
Multimodal data from sensor networks,
Graph signals

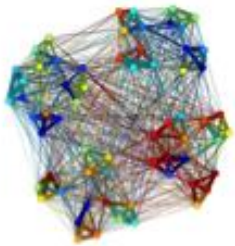




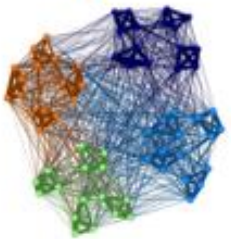


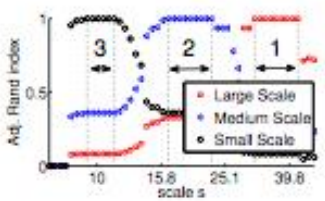




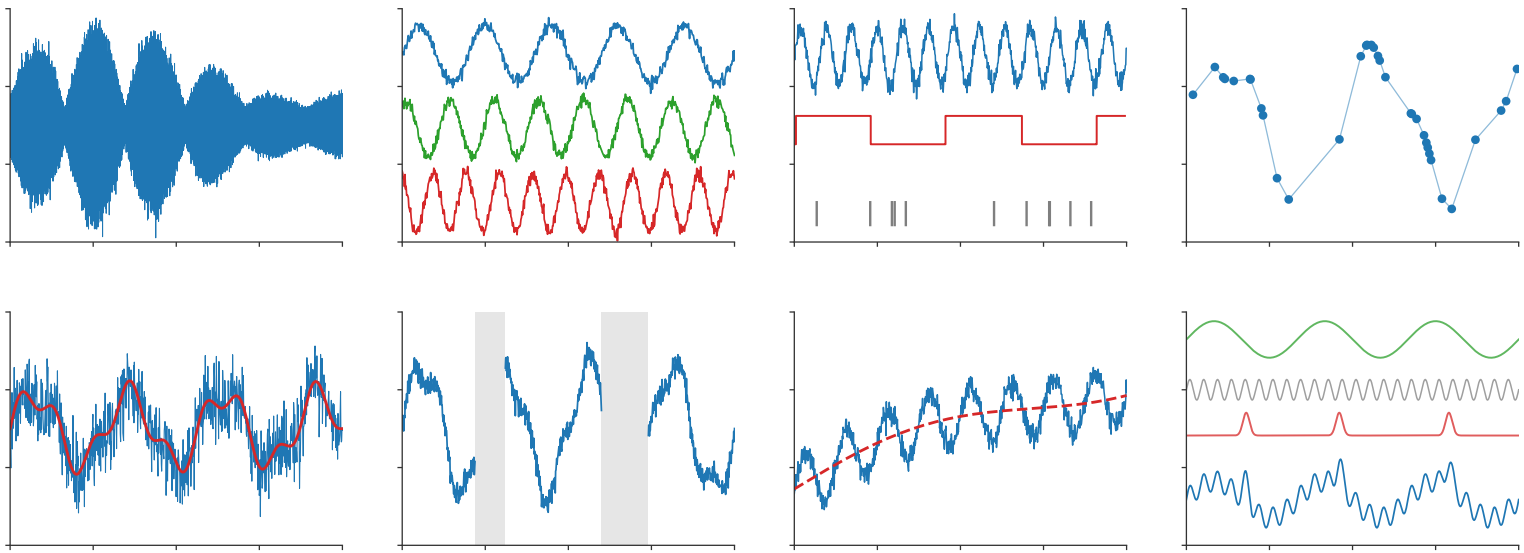








Time series are complex



- Potentially **massive** data (e.g. sound : sampling frequency 44.1 kHz)
- **Multivariate, multimodal, heterogeneous**
- **Noisy, missing data**, trends, mixture of sources

Annotations and ground truth

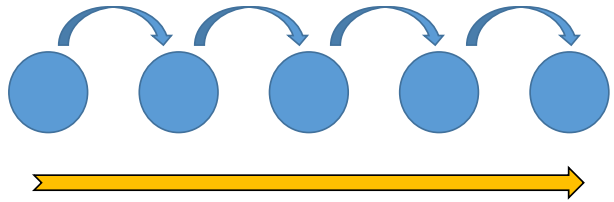
- Often linked to an **application context**: the data scientist is not trained to understand the data
- Contrary to basic image processing tasks (e.g. classification of cats and dogs), annotating time series often **requires expertise**
- **Typical context**:
  - Noisy and dirty data
  - A few annotated signals with blurry labels (confusing and hyper-specialized annotations that cannot be transformed into class labels)
  - An expert with several years in the business, but unable to translate it into ML-compatible annotations

How to use ML in this context?

What about time?

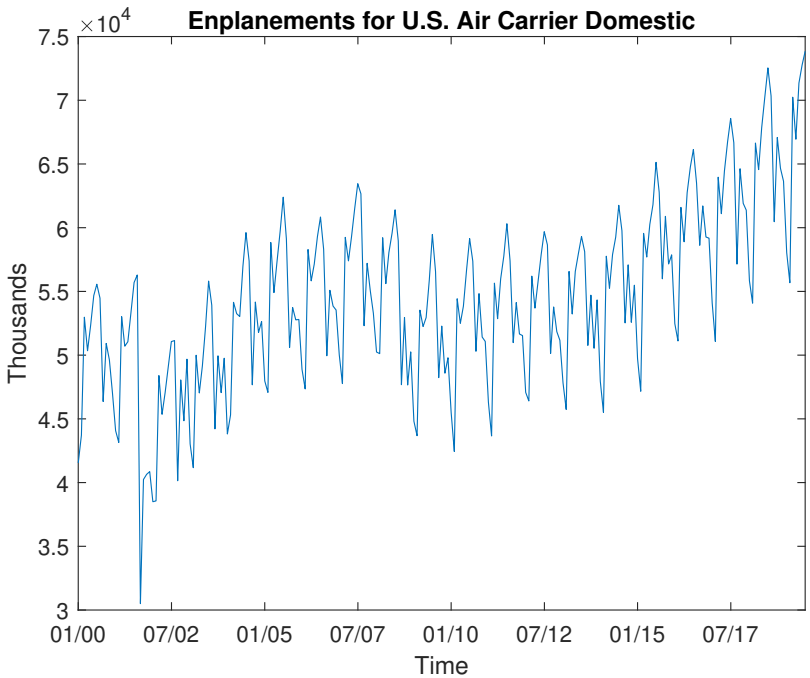
- What is the difference between regular data and time series ? Notion of **sequence and chronology**

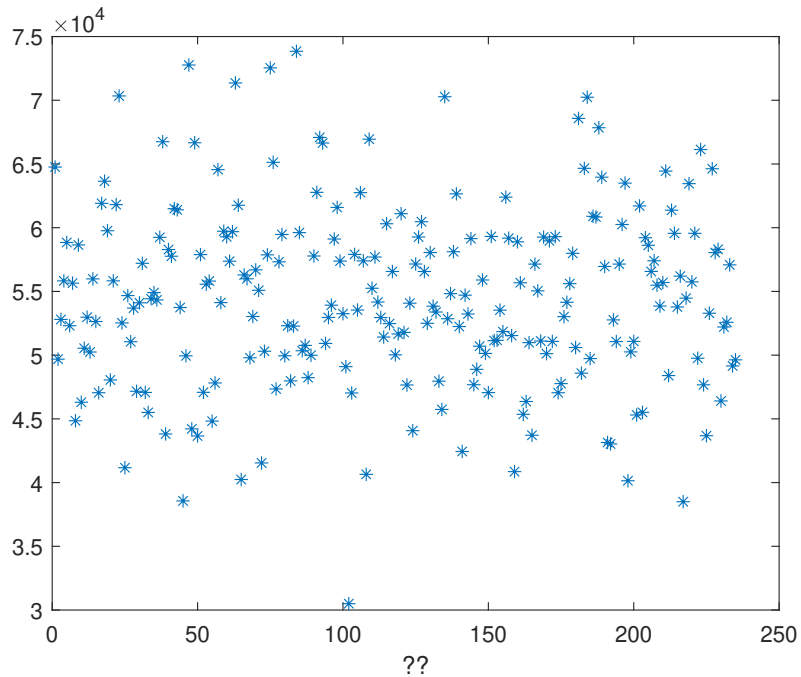




- Each sample corresponds to the measurement of a phenomenon at a given time stamp.
- Time allows us to study the **evolution** of the phenomenon and **should be taken into account** for processing the data

What about time?





Same time series... but mixed up times

World vs. Machine Learning





0.2
0.1
0.4
1.1
0.9
0.3
0.5
0.2

ML

Male
Young
British accent
« Hello »

43
28
21
08
53

ML

New user
Likely to buy a car

- Most ML algorithms **do not care about time**
- How can we still use the time information to extract relevant features/patterns that can be used within an ML procedure ?

Two visions: physics vs. statistics

- The notion of time has been used and modeled in physics since the 18th century and before (e.g. Fourier transform).
  First vision (physics): a time series $x[1:N]$ is the result of the digitization of a physical phenomenon $x(t)$. Physical properties of this phenomenon can be retrieved and analyzed through the study of $x[1:N]$ (and vice versa).
- Randomness can also play a part to model a wider class of signals.
  Second vision (statistics): a time series $x[1:N]$ is a realization of a stochastic process $X[1:N]$. Statistical properties of this phenomenon can be retrieved and analyzed through the study of $x[1:N]$ (and vice versa).

In most cases, both approaches can be combined.

Deep learning: the optimal solution?

Deep learning achieves state-of-the-art results for several tasks (not always true : see next slides) **BUT**...

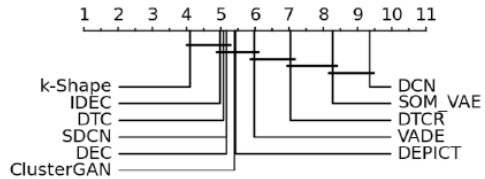
- Good performances $\neq$ good understanding of the data
- DL is a **black box** that may not bring satisfaction to users in the field since they cannot interpret the results
- Early DL methods (e.g. LSTM) only managed at most a few hundred time samples. Long-context architectures exist now, but genuinely **using** distant information remains an **open problem**
- **Data-hungry**: need many labelled examples to avoid overfitting
- **Sensitive to distribution shift**: time series are inherently non-stationary (we will discuss this in depth during the lectures!)
- **Easy to fool**: shortcut learning and adversarial vulnerability

Deep learning: the optimal solution?

To validate our claim, we introduce a set of embarrassingly simple one-layer linear models named LTSF-Linear *for comparison. Experimental results on nine real-life datasets show that* LTSF-Linear *surprisingly outperforms existing sophisticated Transformer-based LTSF models in all cases, and often by a large margin. Moreover, we con-*

Zeng, A., Chen, M., Zhang, L., & Xu, Q. (2022). Are Transformers Effective for Time Series Forecasting?. *arXiv preprint arXiv:2205.13504*.

Forecasting



Shallow (k-Shape) vs. deep

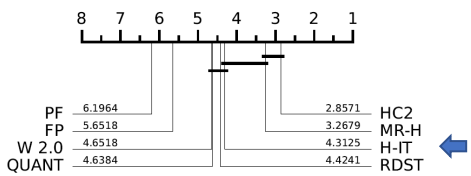
Table 11: Pair-wise comparison of Foundation Models for clustering algorithms with *k*-Shape as the baseline. An asterisk (*) denotes Foundation Models that utilized the UCR dataset in training.

Clustering Algorithm	Better (Adj. P Val)	RI	ARI	NMI	>	=	<
MOMENT*	✘ (1.00e-0)	0.7304	0.2551	0.3436	66	1	61
CHRONOS	✘ (1.00e-0)	0.7172	0.2066	0.2925	47	0	81
OFA	✘ (1.00e-0)	0.7103	0.1949	0.2817	38	0	90
***k*-Shape**	-	**0.7335**	**0.2610**	**0.3444**	-	-	-

Shallow (k-Shape) vs. foundation models

Paparrizos, J., & Bogireddy, S. P. T. R. (2025). Time-series clustering: A comprehensive study of data mining, machine learning, and deep learning methods. Proceedings of the VLDB Endowment, 18(11), 4380-4395.

Clustering



Middlehurst, M., Schäfer, P., & Bagnall, A. (2024). Bake off redux: a review and experimental evaluation of recent time series classification algorithms. Data Mining and Knowledge Discovery, 38(4), 1958-2031.

Classification

In line with related work [67], we found that deep learning approaches are not (yet) competitive despite their higher processing effort on training data. We could also confirm that *"simple methods yield performance almost as good as more sophisticated methods"* [56]. Still, no single algorithm clearly performs best. We highlighted sev-

Schmidl, S., Wenig, P., & Papenbrock, T. (2022). Anomaly detection in time series: a comprehensive evaluation. *Proceedings of the VLDB Endowment*, *15*(9), 1779-1797.

Anomaly detection

= deep learning approach

Deep learning: the optimal solution?

algorithms make a sensible starting point for a new TSC problem. Despite significant research effort, there has not been an Alexnet for TSC, i.e. a deep learning approach that has dominated all others. It may be because the problems in the archives are relatively small compared to other archives used for deep learning evaluation: Table 16 shows that H-Inception time improves relative to other algorithms as the number of training cases increases.

However, we think the core reason deep learning has not provided the gains many expected is that, unlike specific applications such as image classification or natural language processing, there is not one common underlying structure for the neural networks to exploit. Nevertheless, there is no doubt scope for improvements in deep learning algorithms for TSC. H-InceptionTime performs well overall, but Figure 45 demonstrates its limitations: it often performs terribly, and this makes its overall performance worse. If this tendency could be corrected, possibly by some automated structural optimisation, it seems likely that H-InceptionTime could match HC2 and MR-Hydra.

Middlehurst, M., Schäfer, P., & Bagnall, A. (2024). Bake off redux: a review and experimental evaluation of recent time series classification algorithms. Data Mining and Knowledge Discovery, 38(4), 1958-2031.

Classification

5.2.1 Univariate time series forecasting. Table 6 reports the average results of UTSF in terms of the metrics MASE, MSMAPE, and the Ranks in MSMAPE that indicates how many times the best performance is achieved on the datasets. We observe that the recently proposed deep learning methods, including TimesNet [89], PatchTST [64], and N-HiTS [9], exhibit substantially better average performance on univariate datasets in terms of MASE and MSMAPE. However, when considering Ranks, the (non-deep) machine learning methods LinearRegression (LR) [32, 40] and RandomForest (RF) [5] outperform all competitors. This suggests that the machine learning methods may be more suitable in specific scenarios. In this scenario, each individual univariate time series

Qiu, X., Hu, J., Zhou, L., Wu, X., Du, J., Zhang, B., ... & Yang, B. (2024). TFB: Towards Comprehensive and Fair Benchmarking of Time Series Forecasting Methods. Proceedings of the VLDB Endowment, 17(9), 2363-2377.

Forecasting

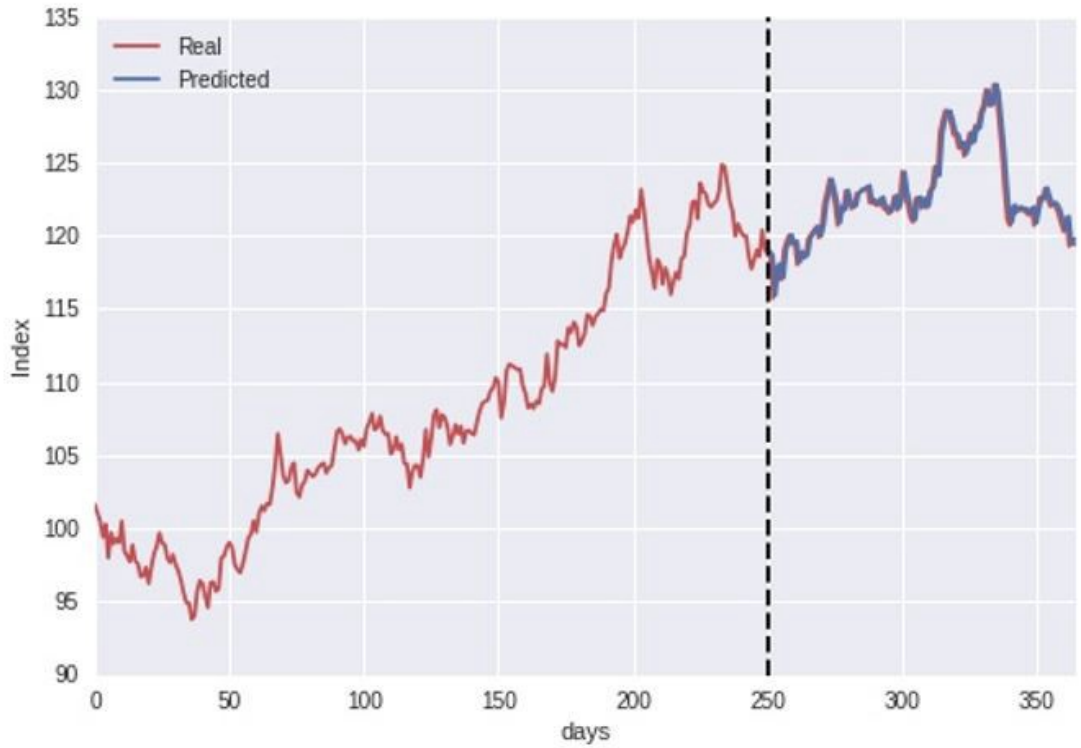
In addition to this line of research, another emerging approach involves prompting LLMs to directly perform anomaly detection tasks. In this setup, an LLM is given a textual prompt (e.g., "Identify anomalies in this time series") along with the corresponding time-series data, expecting it to pinpoint anomalous segments. However, recent studies [31, 171] reveal that while LLMs detect trivial anomalies, they struggle with complex real-world cases, perform better with visual rather than textual inputs, and lack structured reasoning in anomaly detection. Additionally, LLMs' anomaly recognition does not always align with human intuition, detecting obscure patterns while missing obvious anomalies.

Paparrizos, J., Boniol, P., Liu, Q., & Palpanas, T. (2025, August). Advances in time-series anomaly detection: Algorithms, benchmarks, and evaluation measures. In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2 (pp. 6151-6161).

Anomaly detection

How NOT to use DL for time series (1/4)

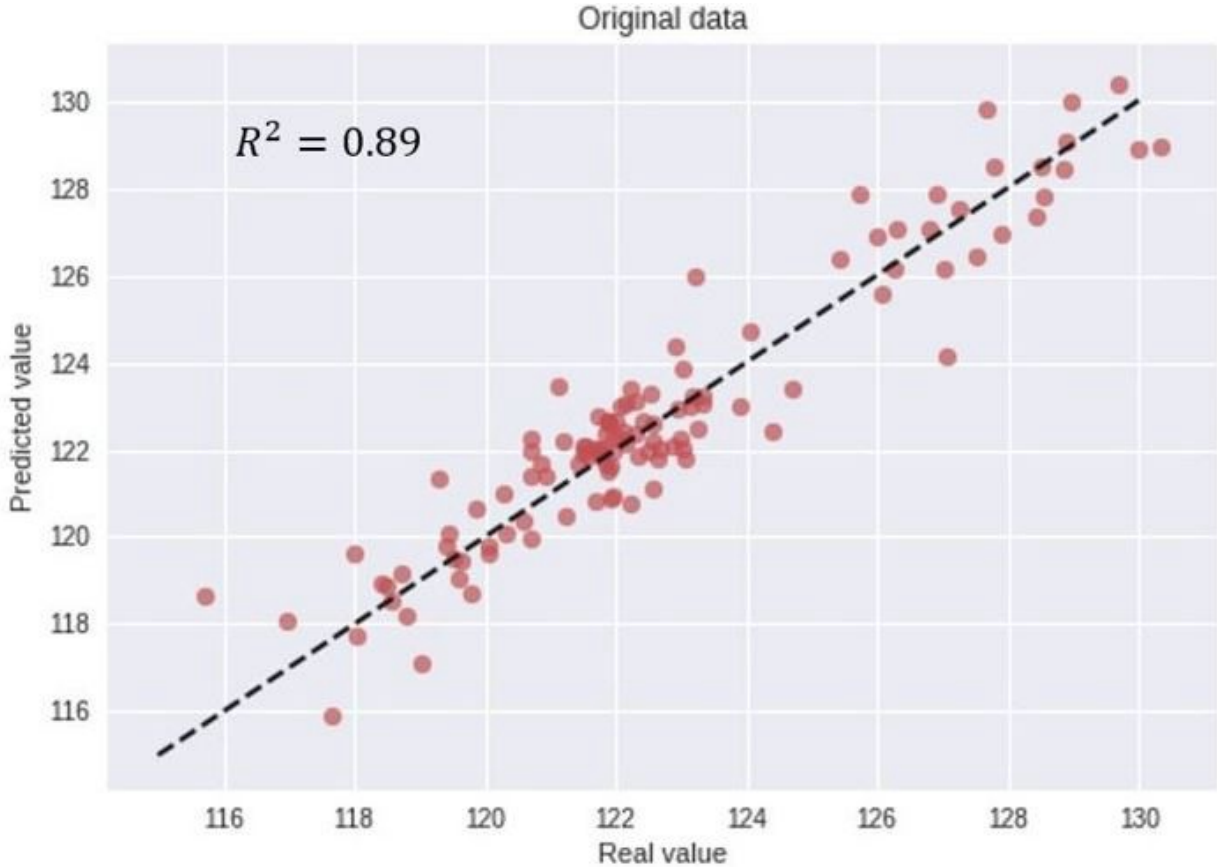
Blog post from V. Flovik
https://towardsdatascience.com/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424



Prediction of stock index with LSTM network: use the first 250 days as training data. Prediction seems great !!

How NOT to use DL for time series (2/4)

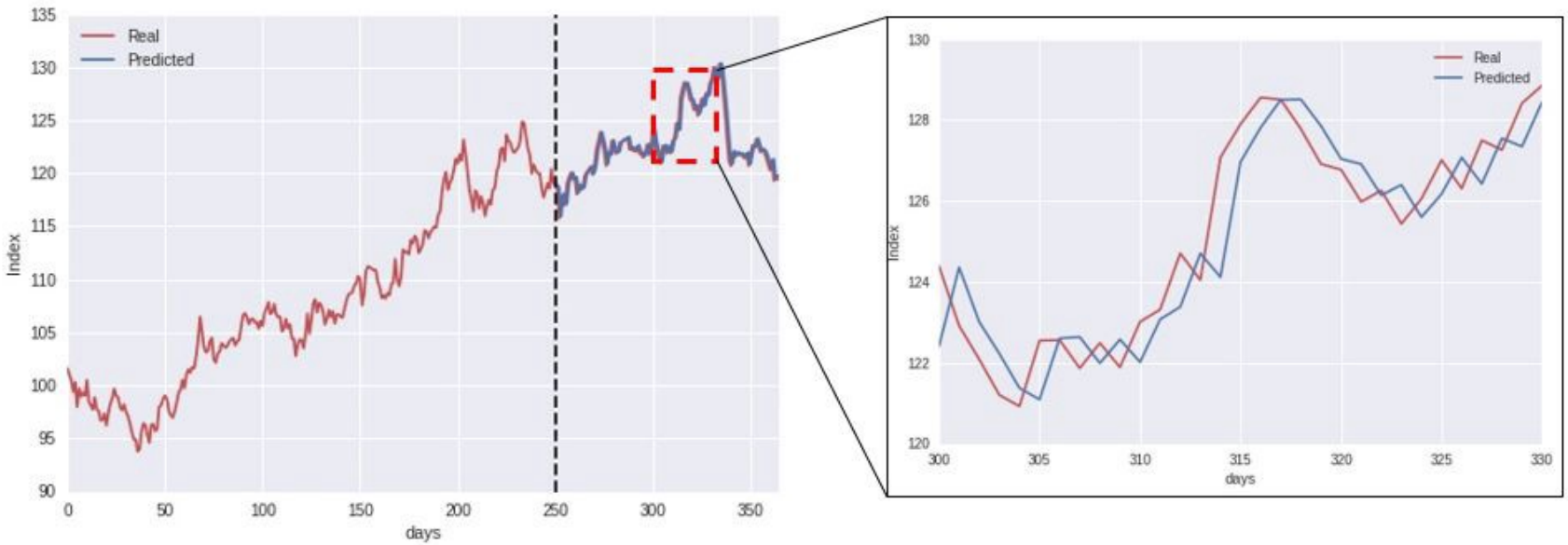
Blog post from V. Flovik
https://towardsdatascience.com/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424



Great performances too! Accurate predictions, low RMSE !

How NOT to use DL for time series (3/4)

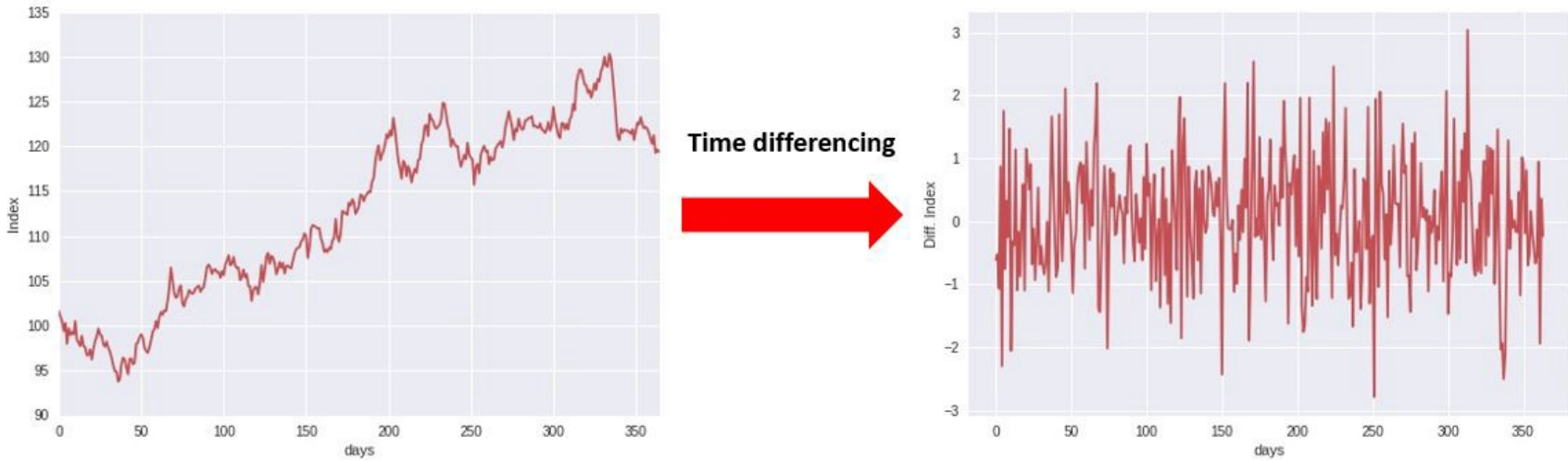
Blog post from V. Flovik
https://towardsdatascience.com/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424



In fact, LSTM was just repeating the previous sample...

How NOT to use DL for time series (4/4)

Blog post from V. Flovik
https://towardsdatascience.com/how-not-to-use-machine-learning-for-time-series-forecasting-avoiding-the-pitfalls-19f9d7adf424



In fact, the data was a random walk : impossible to predict. This could have been detected by a careful pre-investigation...

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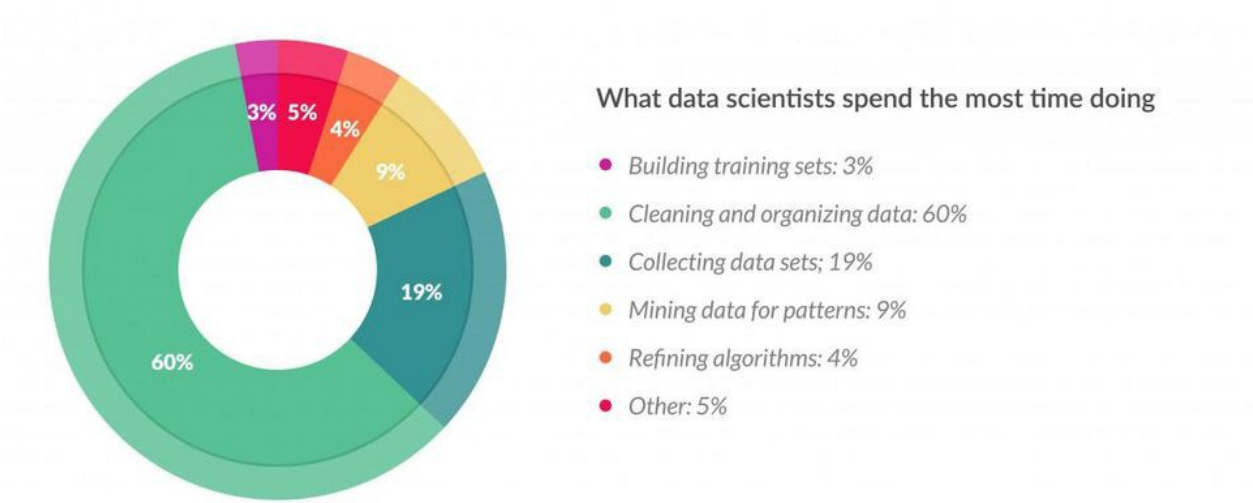
3. Data science for time series

4. Outline of the course

Data science

- Data science is not (or at least **should not be**) attempting to obtain the best performances by launching DL packages in Python
- Data science also aims at **understanding the data**, interacting with experts, bringing human intelligence and expertise and improving knowledge
- In a scarse data context, artificial intelligence cannot be intelligent if the data scientist is not
- Applying complex DL methods does not preclude a thorough preliminary phase... **and ML can also help for this!**

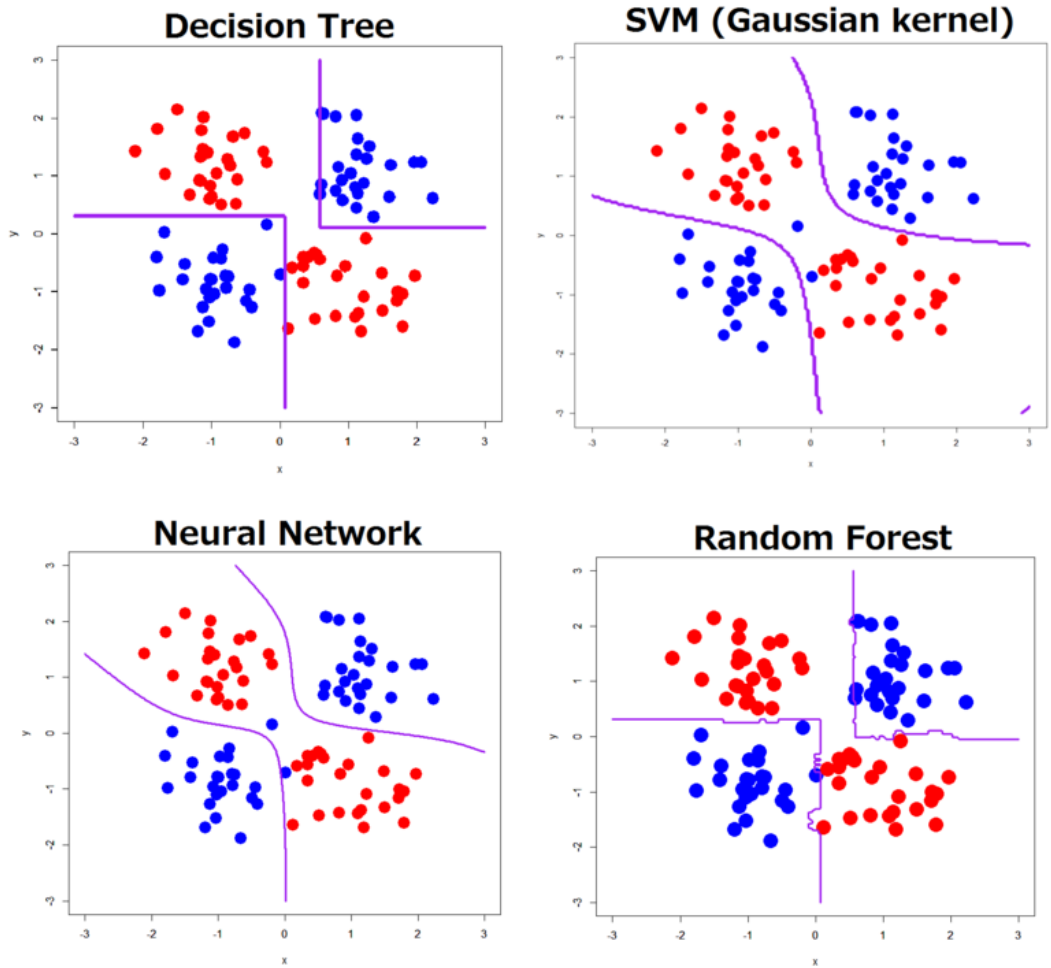
Understanding data: complex and time-consuming task



Source: https://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/

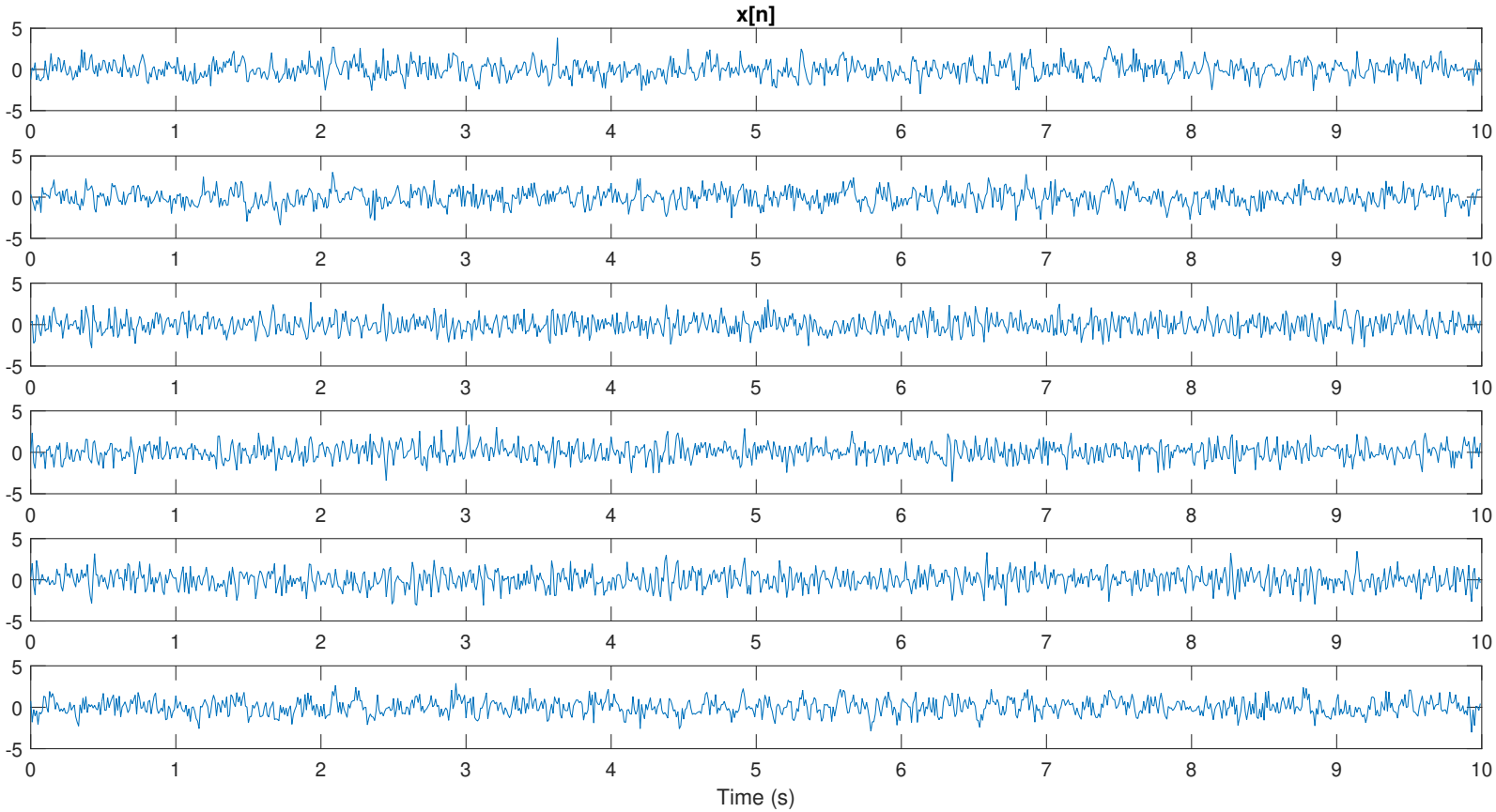
- Understanding the data for extracting the relevant information
- Understand what you do, why you do it and how you do it: **interpretability**

Representation vs. complexity



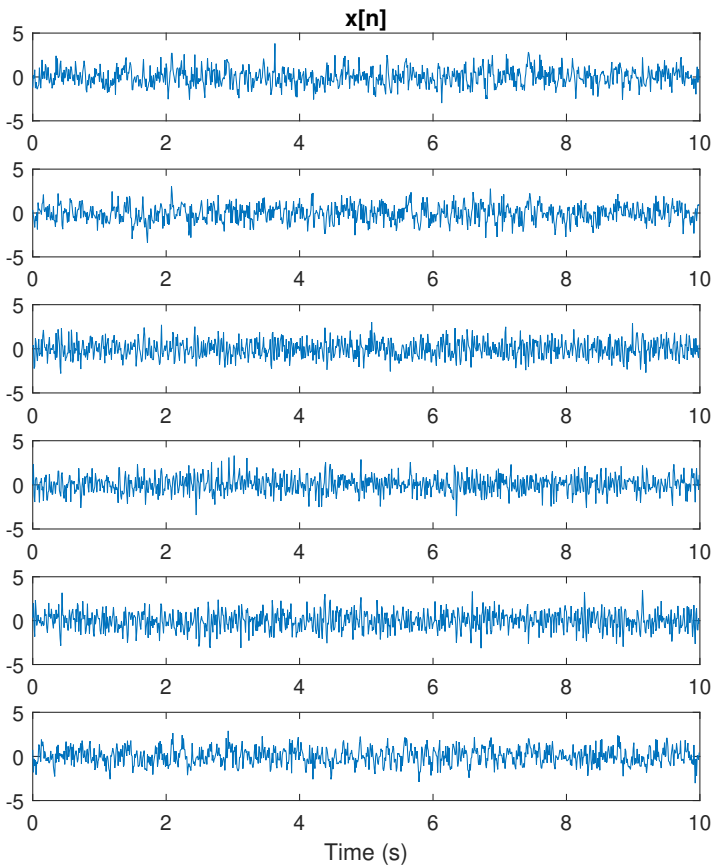
Data complexity often translates into algorithm and model complexity

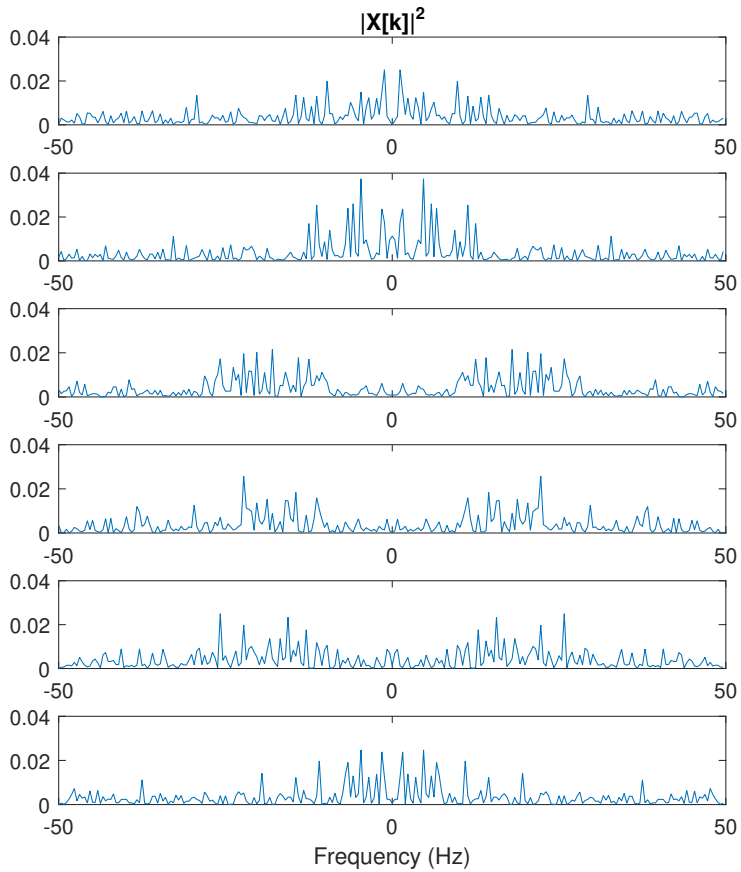
Importance of representation



Two classes of signals?

Importance of representation





Trivial in the frequency domain

Main ML tasks for time series

- **Prediction:** Predict the future values of a time series
- **Completion/interpolation:** Recover missing/lost samples in a time series
- **Classification:** Assign a class label to a time series or to a subsequence
- **Extrinsic regression:** Predict a continuous external variable from a time series (unlike classification, the target is numeric, not a class label)
- **Clustering:** Form several groups of time series with the same properties
- **Query by content/indexation:** Given an input time series, retrieve the closest time series in a large database up to a given measure of fit
- **Prototyping:** Find one or several representative sequences summarizing a set of similar time series
- **Segmentation/change-point detection:** Find significant abrupt changes in the time series
- **Anomaly detection:** Find abnormal events in a time series
- **Pattern extraction:** Find repetitive events in a time series

Hidden ML tasks for time series

- **Understand the data:** know where they come from, how they were acquired, what their characteristics are, interact with domain-experts and understand their problems
- **Improve the data:** find accurate representation spaces where the events of interest can be seen, consolidate the data (denoising, detrending, detection/removal of outliers)
- **Model the data:** physical/statistical or expert-based models, simple, adaptive and interpretable models
- **Extract information from the data:** find repetitive patterns, features of interest, change-points, anomalies

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Aim of the course

Machine Learning for (understanding) Time Series

- Focus on the **hidden tasks**: understand, improve, model and extract information
- Interpretable and reproducible ML algorithms: **white boxes** (no Deep Learning)
- Unsupervised and semi-supervised ML approaches
- Methodology can be applied for prediction, classification, clustering etc...

Outline of the course

- Lecture 1: Pattern Recognition and Detection
- Lecture 2: Feature Extraction and Selection
- Lecture 3: Models and Representation Learning
- Lecture 4: Data Enhancement and Preprocessings
- Lecture 5: Change-Point and Anomaly Detection
- Lecture 6: Multivariate Time Series